CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Pharmacy (B. Pharm)

Semester – II University Backlog Theory Examination May-2015

PH108 Pharmaceutical Chemistry-II(Physical)

Date: 15/05/2015, Friday **Time:10:00 a.m To 01:00 p.m** **Maximum Marks: 80**

Instructions:

1. **Figures to right indicate marks.**
2. **Section wise separate answer book is to be used.**
3. **Draw neat sketches wherever necessary.**

SECTION- I

Q 1		**Attempt any TWO of the following:**	
A		What is surface tension? Write about various methods for the estimation of surface tension.	**10**
B		Explain following terms: [1] Refractive index, [2] Parachor, [3] Dipole moment,[4] Optical activity, [5] Viscosity.	**10**
C		Compare properties of radioactive rays. Explain the diagnostic applications of radiopharmaceuticals.	**10**
Q 2		**Attempt any TWO of the following:**	
A		Explain in detail about Jablonski diagram and state beer's law of photometry.	**10**
B		Give characteristics of enzyme catalysis.Write a short note on Acid-Base catalysis.	**10**
C		Define Quantum efficiency. Discuss causes of high quantum yield and low quantum yield with example.	**10**

SECTION- II

Q 3		**Attempt any TWO of the following:**	
A		What is mean by Phase rule? Write a brief note on One component three phase system in detail.	**10**
B	**1.**	What is conductance? Discuss Debye-Huckel theory.	**05**
	2.	Write a note on Joule-Thomson effect.	**05**
C		Write a note on Carnot cycle and derive equation of thermodynamic efficiency.	**10**
Q 4		**Attempt any TWO of the following:**	
A		Explain: Calcium as an important electrolyte of body.	**10**
B	**1.**	State and explain first law of thermodynamics.	**05**
	2.	Importance of Copper as an electrolyte.	**05**
C	**1.**	What is partition coefficient? How it can be practically measured?	**05**
	2.	Differentiate Ideal Solution and Real solution in detail.	**05**